

Teaching Statement

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Teaching Philosophy: Mathematics of Real Systems

I came to research through wireless engineering, which taught me that the best teaching does not separate theory from practice. I now teach the same way I do research: by introducing students to a real system, helping them discover its structural invariants, and only then formalizing those invariants mathematically. Students who learn distributed systems by deriving CAP-theorem constraints from a working key-value store retain the theory in a way that students who memorize the theorem alone do not.

This philosophy is the same one that animates my research, predictive adaptation with formal guarantees, applied to pedagogy. A student who can extract structural invariants from a real protocol has acquired a transferable skill; a student who has only memorized a theorem has not. The challenge of teaching wireless networks, distributed systems, or any infrastructure subject is to give students enough hands-on contact with real systems to develop intuition while preserving the formal scaffolding that lets them reason rigorously when intuition fails.

I have tested this philosophy in two graduate-level courses I designed and solo-taught at the University of Sistan and Baluchestan (USB), Iran, in Fall 2022 and Spring 2023.

Teaching Record

Computer Networks (USB, Fall 2022 and Spring 2023). Solo instructor, graduate level, 50+ students per offering, hybrid synchronous and asynchronous delivery. I designed the syllabus from scratch, building from the physical layer (modulation, coding, channel access) through the network layer (IP routing, congestion control) to the application layer (HTTP, QUIC, BBR). I integrated material from my own research, the Wi-Fi MU-MIMO scheduling work from my Skylark internship; the CellDAM cellular-detection work from my dissertation; the cache-coordination work that became CacheCatalyst, as case studies that let students see how recent research engages real protocols rather than toy models.

Advanced Wireless Networking (USB, Fall 2022 and Spring 2023). Solo instructor, graduate level, 50+ students per offering, hybrid synchronous and asynchronous delivery. This course explicitly aimed to bridge the gap between standard-track wireless courses (which often present Wi-Fi and cellular as separate, settled systems) and current research (where Wi-Fi 7 multi-link operation, 3GPP NR, NTN, and direct-to-device satellite are converging into a single hierarchical wireless substrate). I structured the course around three substrates, terrestrial Wi-Fi, terrestrial cellular, and satellite, and showed students how the same fundamental questions (channel access, mobility management, security) appear differently in each.

These courses gave me concrete experience with three pedagogical challenges that any future course of mine will face. First, students arrive with deeply uneven preparation. Some have strong information-theory background but no implementation experience; others have built and debugged wireless drivers but cannot derive Shannon's theorem. I learned to design problem sets that have two entry points, an analytical one and an implementation one, that converge on the same insight. Second, the asynchronous component of hybrid teaching is not just video lectures; it requires asyn-

chronous exercises with feedback that does not require synchronous instructor response. I learned to use auto-graded coding exercises (in Python and ns-3) for the formative components and reserved synchronous time for higher-order questions. Third, students engage more deeply when the course material connects to current research. I made every lecture include at least one open research question with a recent paper as starting point, and required each student to produce a 4-page literature analysis on a research topic of their choice as the term project.

Pedagogical Innovations I Will Bring

Lab-first instruction in wireless and networked-systems courses. The single largest barrier to learning wireless and networking is that students never see real systems. A standard wireless course teaches modulation in MATLAB without ever touching an SDR; a standard networking course teaches TCP without ever opening Wireshark. I have access to a teaching lab equipped with USRP B210s, ESP32 boards, Open5GS / srsRAN cellular stacks, and StarryNet satellite emulation, and I will design lab sequences that put students in front of these systems in week 2. The Skylark MU-MIMO scheduler I helped design started as a graduate-student lab exercise on real Argos basestations; that pedagogical pattern, from real-hardware encounter to working algorithm to publication, is replicable.

LLM-assisted protocol comprehension. The volume of normative specification text in modern protocols (3GPP TR/TS, 802.11be, RFCs) has outpaced the standard graduate-course reading budget. My research has shown that LLMs can extract typed primitives from these documents in a way that supports student learning, not just automated detector synthesis. I will develop a course module, “Reading the Standards”, in which students use LLMs as protocol-comprehension assistants to read sections of 3GPP TS 33.501 or 802.11be that they would otherwise skip, then verify the extracted primitives against open-source implementations.

Real-data course projects from production traces. My research provides large-scale, real, labeled traces that are difficult to obtain elsewhere: 106 GB of 5G commercial carrier traces from CellForge, 73.4M Doppler observations from the satellite portfolio, and the BlackWidow 47-node testbed dataset. These can be released to students under appropriate anonymization for course projects, giving them the chance to do real research on real data within a semester.

Cascade-safety as an undergraduate distributed-systems topic. The intuition that cascading failures in distributed systems behave like epidemics, with an analog of the SIR reproduction number, is teachable at the undergraduate level. It also connects undergraduate systems courses to ongoing graduate-level research in safe reinforcement learning. I will develop teaching material that introduces R_0 -based cascade analysis to distributed-systems undergraduates as an alternative to the standard “the network goes down and that’s bad” framing.

Courses I Am Prepared to Teach

At the undergraduate level: Computer Networks, Operating Systems, Distributed Systems, Wireless Communication, Introduction to Computer Security and Privacy.

At the graduate level: Advanced Wireless Networking, Cellular and 5G Systems, Datacenter Networking, Satellite and Space Systems, Mobile and Edge Computing, Machine Learning for Systems, Safe Reinforcement Learning, Privacy in Wireless and IoT Systems.

New seminar courses I would develop: “LLM Systems and Inference Infrastructure”; “Direct-to-Device Satellite Networks”; “Formal Methods for Wireless Protocols”; “The Modern Datacenter:

From L4 Load Balancing to Carbon-Aware Scheduling.”

Mentorship and Student Development

I have co-supervised two PhD students at NC A&T (Mahsa Tavasoli on vehicular cloud-edge offloading; Radwan Rouzky on data governance in smart agriculture). I mentored Mohammad Sepahi (master’s student, Old Dominion University; now a research engineer at Rivian) through his thesis and a USENIX VehiSec ’26 publication as first co-author. I mentored Alexander Florez (INSA / Inria-Lyon) to a Best Paper Award for the ML in Satellite IoT survey. I have served as a research mentor for three undergraduate students at Old Dominion University working on edge ML, data quality, and healthcare ML.

Each of these mentorships shares a structure: I work with the student to identify the structural invariant of the problem they are working on, what is the actual scientific question, beneath the engineering challenge they think they are solving?, and then we trace that invariant through to the right experimental design. This approach helps undergraduate and master’s students transition from problem-solving to problem-formulation, which is the actual hard transition in becoming a researcher.

I am committed to recruiting and mentoring students from underrepresented backgrounds, building on my own experience as a woman from Sistan and Baluchestan, Iran’s least-developed province, who progressed through three educational systems (Iranian, French, American) to a research career at UC Berkeley. The detail of that commitment is the subject of my diversity statement.

Why Teaching Matters to My Research Vision

I believe the most consequential research groups are the ones that produce both papers and people. The CellForge, BlackWidow, SkyCascade, and Stellink lines of work that anchor my research depend on students who can read normative specification text rigorously, build and debug wireless testbeds, derive cascade dynamics from first principles, and integrate ML methodology into systems work. There is no shortcut to producing such students; only sustained teaching that does the slow work of moving them from operator-level understanding (“the protocol does X”) to researcher-level understanding (“the protocol does X because of structural property Y, which we could relax in setting Z”). My research agenda and my teaching agenda are the same agenda: building a generation of researchers who can extract invariants from real systems and turn them into the next generation of predictive, verifiable infrastructure.